

TEEP RESEARCH PROJECT B

Cuffless Blood Pressure Estimation

The goal, the two datasets, and how the data was prepared

Rachit Ranka

Every number here was measured from the data, not quoted from memory

What we are doing, and what we are checking

Blood pressure needs a cuff. A watch already has a sensor that might replace it.

A cuff squeezes your arm to measure blood pressure. It works, but an elderly person living alone cannot manage it five times a day. A smartwatch already shines a light into the skin and records the pulse, all day, for free. That signal is called PPG. Can a computer read blood pressure out of it?

Many papers say yes. But a lot of them tested the wrong way: they put some clips from a person in the training set and other clips from the SAME person in the test set. The model had already met that person. It is like giving a student the exam paper the night before — the marks tell you nothing.

1. How much better does that make the results look?
2. What is the real number, on a person the model has never met?
3. What happens on a real smartwatch instead of hospital equipment?
4. If we give the model a few real readings from the new person, how much does it improve?

Two datasets

One to teach the model. One to check whether it survives outside the hospital.

	PulseDB	PPG-DaLiA
Where it comes from	Hospital ICU and operating rooms	Ordinary daily life
Sensor	Clinical finger sensor + ECG	Smartwatch on the wrist
People	1,553 patients	15 volunteers
Recording per person	1 hour (60 to 100 min, by group)	2.4 hours (88 to 177 min)
Total	1,794 hours · 646,000 clips of 10 s	36 hours
Blood pressure labels	Yes — measured, one per clip	None at all
We use it to	Train the model and test it	Check it on real-world data

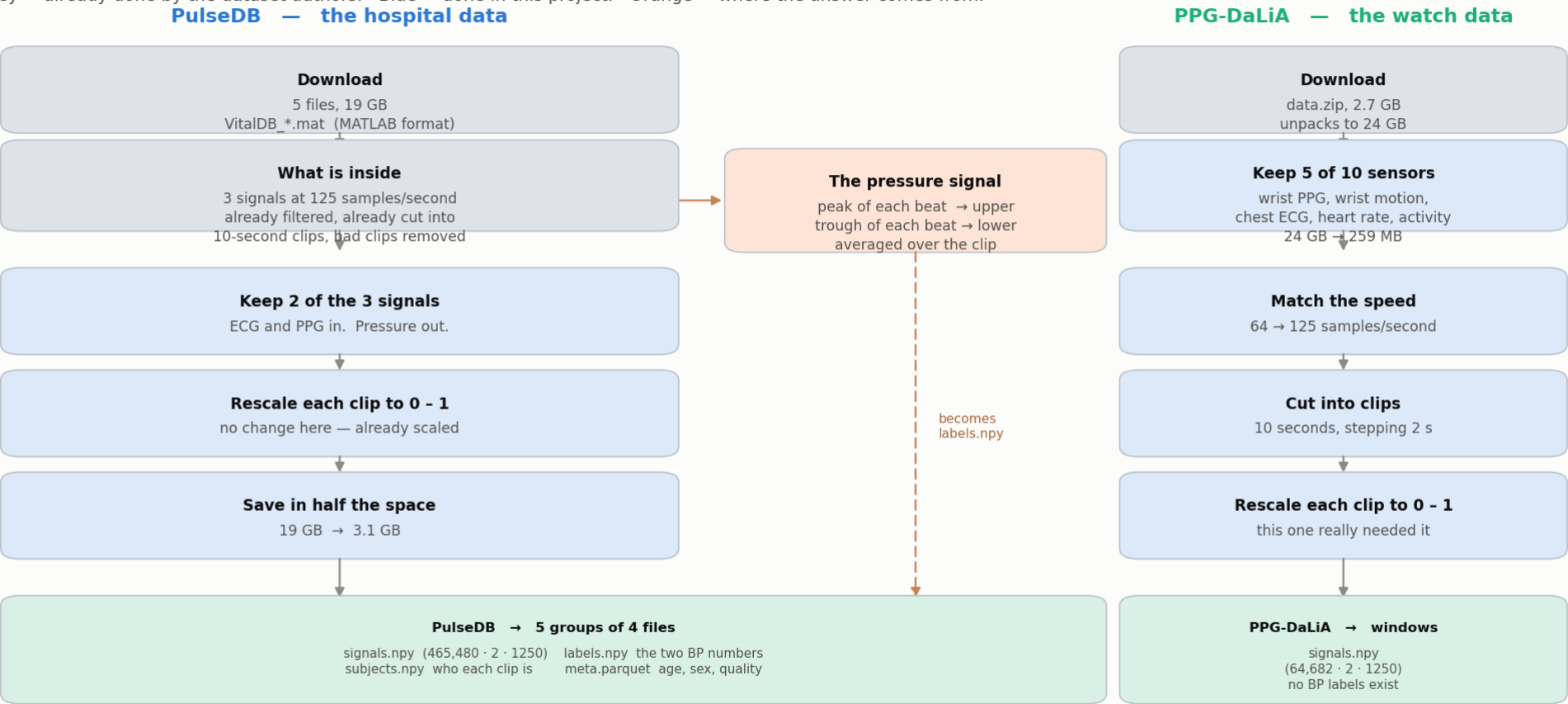
PulseDB is Boston (MIMIC-III) and Seoul (VitalDB) combined, 5,361 patients. The spec says to use the Seoul half only — hence the “VitalDB” filenames. It ships already split into five groups, and every published paper on PulseDB uses those same groups.

Has anyone used both datasets together? One paper has (arXiv:2508.10805, 2025), but for removing motion noise and estimating heart rate — never blood pressure. The pairing exists; using it for blood pressure does not.

The whole pipeline on one page

How the raw downloads become the files a model reads

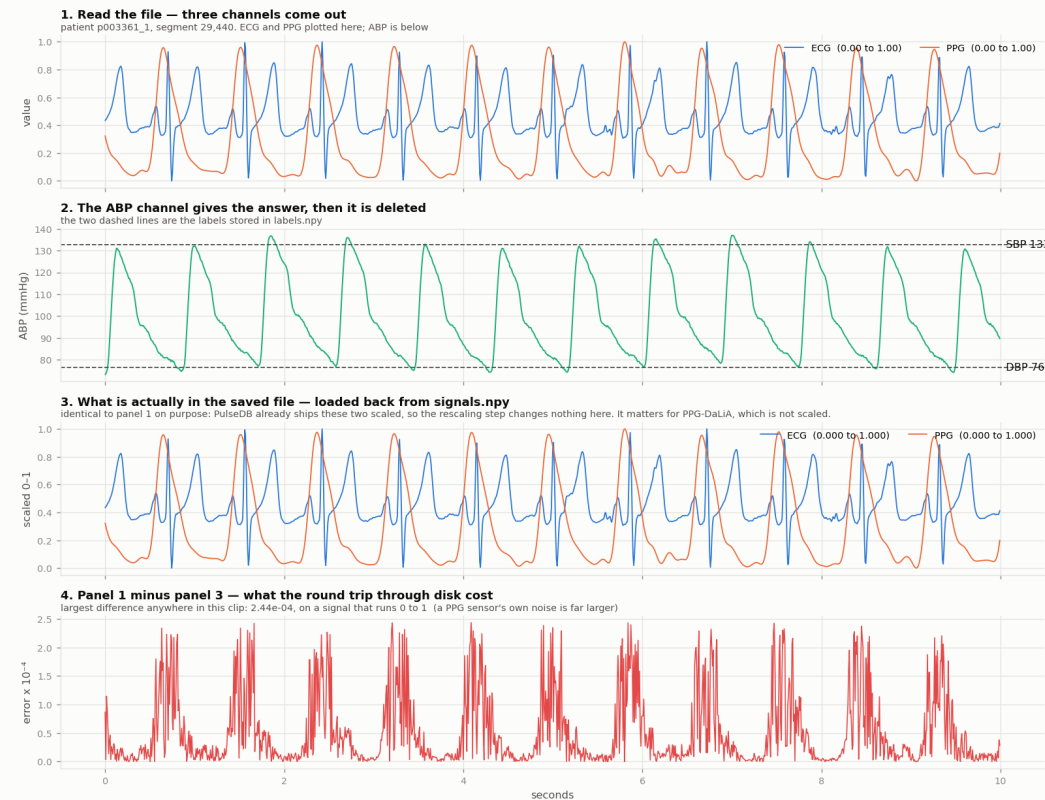
Grey = already done by the dataset authors. Blue = done in this project. Orange = where the answer comes from.



Both end in exactly the same shape — 2 signals of 1250 numbers per clip — so a model trained on the hospital data runs on the watch data without a single change. That is the only reason the real-world test means anything. Last step, not drawn: patients are split into groups and the code checks that none appears in two of them.

One real clip, every step

Patient p003361_1, one 10-second clip, from file to stored array.

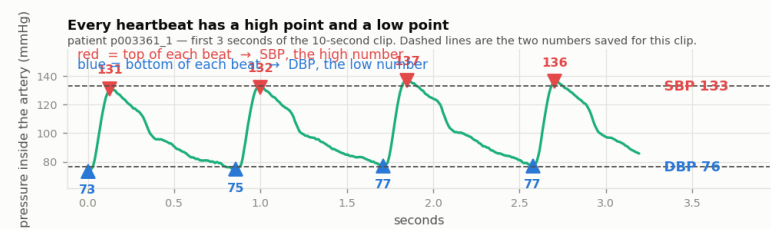


Panel 2 is the important one — the green trace produced this clip's answer of 133 over 76, and is then thrown away. The model only ever sees the blue and orange lines.

Panel 3 looks identical to panel 1 because PulseDB already scales these two channels. That step exists for PPG-DaLiA, which does not.

Where the two numbers come from

A blood pressure reading is the top and the bottom of one wave.



Each red triangle is one heartbeat squeezing.
Each blue triangle is the heart relaxing before the next beat.

That is all blood pressure is: the high number and the low number of the same wave.

OVER THE WHOLE 10-SECOND CLIP

heartbeats found	12
average of the high points	133.4 mmHg
average of the low points	75.6 mmHg
label saved as SBP	132.8 mmHg
label saved as DBP	76.4 mmHg

Our averages land within about 1 mmHg of the saved labels. They are not identical, and that is expected: PulseDB used a published beat-detection algorithm (Elgendi's) to place the peaks, and this figure re-finds them with a simpler method. The labels used everywhere in this project are PulseDB's, not these.

The picture on the left shows only the first 3 seconds, so counting those triangles alone gives a different average again.

A thin tube in the patient's artery measured this directly. Nobody estimated it.

That is why the model is never shown this green line - it would be handing over the answer sheet.

Red triangle = the heart squeezing, the high number. Blue = the heart relaxing, the low number. Average them across the clip and you get 133 over 76.

A thin tube inside the patient's artery measured this. Nobody estimated it — which is why these labels can be trusted, and why the model never sees the green line.

The two datasets arrive in very different states

PulseDB comes finished. PPG-DaLiA does not — which is why it needed more work.

Step	PulseDB	PPG-DaLiA	So who did it?
Sensors synchronised	done by authors	done by authors	neither of us
Signal filtered	done — Chebyshev-II, 0.5–8 Hz	not done	nobody — see note
Cut into fixed clips	done — 10 s, non-overlapping	not done	we did, for DaLiA
Bad clips removed	done — three quality rules	not done	we score quality, keep all
Signal rescaled	done — 0 to 1	not done	we did, for DaLiA
Sampling rate	125 Hz	64 Hz wrist, 700 Hz chest	we resampled DaLiA to 125
Blood pressure labels	yes — from an arterial line	none at all	cannot be created

PulseDB's authors shipped the data already filtered, segmented, quality-checked and labelled, so we left it exactly as it came. Filtering an already filtered signal removes real detail rather than cleaning it.

PPG-DaLiA's watch does its own processing on the device and the authors ship what it produced, so we did not add a filter there either — the real-world test only means something if both are treated identically. Its readme suggests 8-second clips; we use 10 so the shape matches PulseDB.

What we ended up with

Group	Clips	Patients	Per patient	What it is for	Size
train	465,480	1,293	60 min	The model learns from this	2.33 GB
calfree_test	57,600	144	67 min	The real exam — brand new patients	288 MB
aami_cal	70,212	116	101 min	A few readings to tune the model	351 MB
aami_test	666	116	1 min	Then score it on those same people	3 MB
calbased_test	51,720	1,293	7 min	Easy case — patients it knows	259 MB
PPG-DaLiA	64,682	15	144 min	Watch data, no BP labels	323 MB

Each group is four files: the waveforms, the two BP numbers, which patient each clip came from, and a small table of age, sex, weight and signal quality. They line up row by row.

One thing not to misread: PPG-DaLiA's clips are cut every 2 seconds and are 10 seconds long, so they overlap. Its recording really is 36 hours in total, not 180.

Three things that could have gone wrong

Each one was tested, not assumed.

What could go wrong	How we checked	What came back
The saved file might not match the original	Reloaded four clips from the download and compared them number by number	Same to within 0.00024 — that is 0.02%
Saving in a smaller format might have damaged the signal	Ran pulse detection on both versions and compared where it found the heartbeats	Identical on all 300 clips tested
A patient used for training might turn up in a test group	Compared the patient lists of every pair of groups	Zero shared, and it is re-checked on every run

The third one matters most. If a training patient ever appeared in a test group, the results would quietly get better and nobody would notice — so the build stops instead.

Two groups do share patients on purpose (aami_cal with aami_test, train with calbased_test). Both are by design, and neither is ever a headline result.

Did we prepare the data the same way as the papers?

Our settings against the two most relevant studies, and how much data each had.

Setting	PulseDB gives	Benchmark 2025	DMT 2026	Us	Same?
Measurements per second	125	125	125	125	Yes
Clip length	10 s / 1250	10 s / 1250	1250	10 s / 1250	Yes
Cleaning / filtering	already done	kept as is	not stated	kept as is	Yes
Patient groups	3 official ones	official	official	official	Yes
Input signals	ECG + PPG	PPG only	PPG only	ECG + PPG	No
Rescaling	0 to 1	not stated	z-score	0 to 1	No
Training clips	—	418,986	465,480	465,480	Yes

Four of six settings match exactly, because they come with the dataset and nobody can change them. Our preprocessing is the standard preprocessing for this dataset, not something we invented.

The two that differ are deliberate: we keep the ECG because the spec asks for it, and we use the 0-to-1 scaling the dataset already provides rather than adding another step.

The last row matters: the best published result trained on 465,480 clips. We hold exactly the same number.

Where this leaves us

The data is ready. Nothing has been modelled yet.

Both datasets are downloaded, converted, checked, and stored in a form a model can train on.

- 1,553 patients from PulseDB, plus 15 volunteers from PPG-DaLiA

- 1,794 hours of hospital recording and 36 hours of watch recording

- 3.6 GB on disk, down from 22 GB of downloads

- no training patient appears in any test group, checked on every run

Clips from both datasets now have exactly the same shape, so a model trained on hospital data can be run on the watch data without a single change.

Everything regenerates from scripts — delete the processed folder and one command rebuilds it.

Next: train the first models, then compare against the published results on the same dataset.

References

THE TWO DATASETS

PulseDB: a large, cleaned dataset based on MIMIC-III and VitalDB — Wang et al., Frontiers in Digital Health, 2023.

doi.org/10.3389/fdgth.2022.1090854

PPG-DaLiA — Reiss et al., UCI Machine Learning Repository, dataset 495.

THE ONE PAPER USING BOTH OF THEM

— for denoising and heart rate, not blood pressure

Reduction of motion artifacts from PPG signals using learned convolutional sparse coding, Physiological Measurement, 2025. [arXiv:2508.10805](https://arxiv.org/abs/2508.10805)

PAPERS THAT PREPARE THE SAME DATA THE SAME WAY

Generalizable deep learning for PPG-based blood pressure estimation — a benchmarking study, 2025. [arXiv:2502.19167](https://arxiv.org/abs/2502.19167)

DMT: Demographic Conditioning, Morphology-Enhanced Transformer, 2026. [arXiv:2606.11125](https://arxiv.org/abs/2606.11125)

Validation of deep learning models for cuffless BP estimation, 2024. oaepublish.com/articles/chatmed.2023.23

rU-Net, Multi-Scale Feature Fusion and Transfer Learning, IEEE JBHI, 2024.

WHY THE SPLIT MATTERS

Investigation of Data Leakage in Deep-Learning-Based Blood Pressure Estimation, IEEE, 2023.